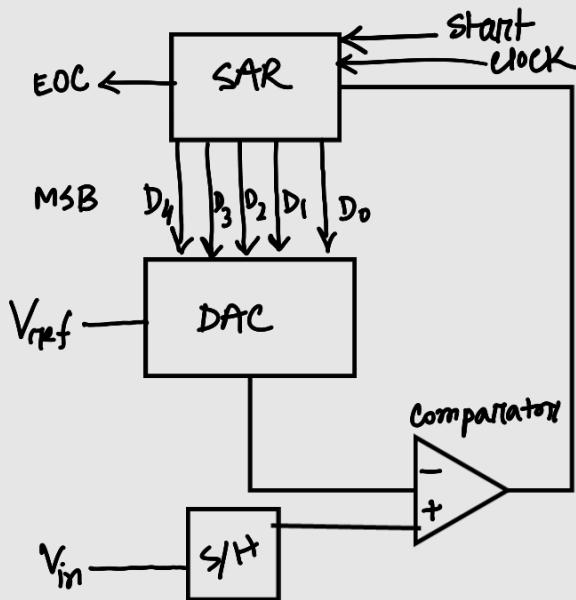


2. a) Sketch the block diagram of a 5-bit successive approximation A/D converter. For this 5-bit successive approximation A/D converter; assume that the reference voltage applied to the D/A converter circuit is $V_{REF} = 10V$. For a particular instant of time; if an analog input voltage has the magnitude of $V_{IN} = 6V$, determine what will be the corresponding output bit pattern of the A/D converter after completing the conversion process? (Show the changes in the output bit pattern after each cycle of the conversion process)

15
(CO2,
PO2)



$V_{ref} = 10V$ $N=5$ (5 bits) , so quantization $1/15 = 2^5 = 32$

$$q = \frac{10-0}{32} V = 0.3125$$

D_4	D_3	D_2	D_1	D_0	V_{DA}
0	0	0	0	0	0.00
0	0	0	0	1	0.3125
0	0	0	1	0	0.625
					$\vdots$

$$0 \rightarrow 0.00$$

$$1 \rightarrow 0.00 + 1 \times 0.3125$$

$$2 \rightarrow 0.00 + 2 \times 0.3125$$

$\vdots$

$$N \rightarrow 0.00 + N \times 0.3125$$

$$\text{max } N = 31$$

$$V_{in} = 6V$$

Start from MSB $\rightarrow 10000$ (16)

$$6V > 0.3125 \times 16$$

Next step $\rightarrow$ if less than 01000
if more than 11000

11000 chosen

$$6V < 7.5V$$

So, next 10100 chosen

$$6V < 6.25V$$

So next 10010 chosen

$$6V > 5.625V$$

So next 10011 chosen

$$6V \approx 5.9375V$$

Result $\rightarrow 10011$

[Start signal]

$$\hookrightarrow D_4 = 1, D_3 \dots D_0 = 0$$

Sent to DAC



to Comparator



with analog value
converted using
S/H

When less, $D_4 = 1$
needs to be cleared -
next significant bit

$$D_3 = 1$$

comparator gives high pulse
when higher, then value
can be kept

Next significant bit
again changed,
comparator gives
pulses until D_0 reached

EOC signal sent
(End of conversion)

c) For a certain 2-bit successive-approximation ADC, the maximum ladder output is +8V. If a constant +6 V is applied to the analog input, determine the sequence of binary states for the SAR. Repeat the mentioned problem for a 4-bit successive-approximation ADC.

[8]
CO6,
PO1

2 bit

$$\text{Levels} = 2^2 = 4$$

$$\text{Step size} = \frac{8}{4} = 2$$

0, 2, 4, 6
↙ ↓ ↓ ↓
00 01 10 11

Input +6V $\rightarrow$ SAR $\Rightarrow$ 10

6 > 4 so, move to 11, keeping
MSB = 1

6 = 6 equal. so LSB 1 kept

$\therefore$ 11 is the required code

For 4 bit similar

1. a) Draw the circuit diagram of a 6-bit successive approximation ADC. Consider the clock rate is 3 MHz and the range of the DAC output is $0 \text{ V} \leq V_{DAC} \leq V_{ref}$ in addition to the offset of +0.5 LSB. 10

- b) Calculate the conversion rate (in MHz) of the ADC from question 1(a). 7

6 bit successive approximation ADC

→ clock rate $f_{CLK} = 3 \text{ MHz}$

$$T_{CLK} = \frac{1}{3 \times 10^6} \text{ s}$$

6 bit SAR needs 6 bit trial clock cycle

clocks per conversion = 6

$$\text{conversion time} = \frac{6}{3 \times 10^6}$$

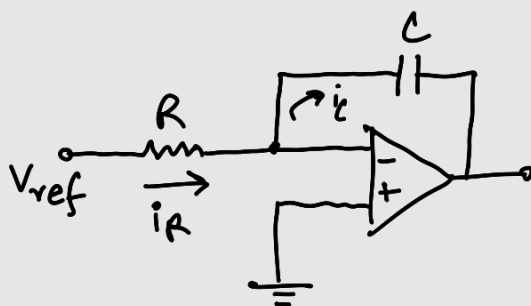
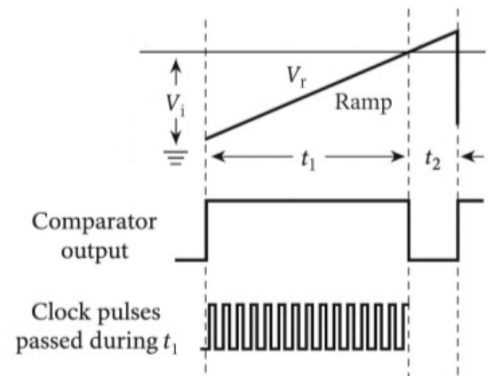
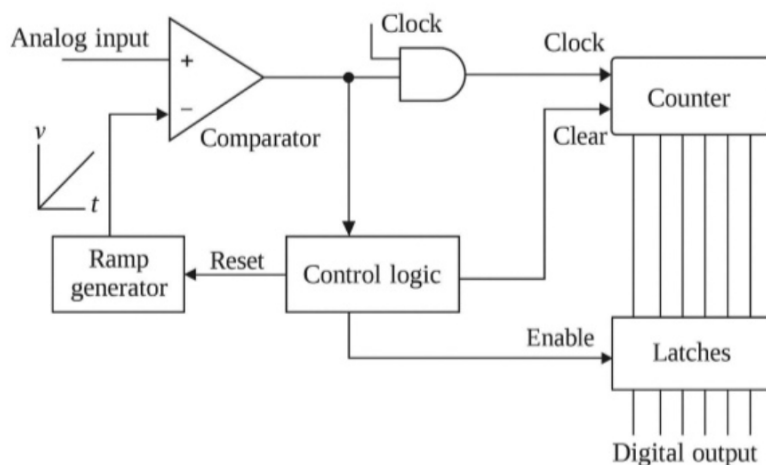
$$\text{conversion rate} = \frac{3 \times 10^6}{6} = 500000 \text{ Hz} \\ = 0.5 \text{ MHz}$$

- b) With proper circuit diagram, derive the expression of total conversion time of a dual slope A/D converter. Consider that a 1 MHz clock generator and a 5-bit counter are used in a dual slope A/D converter. The reference voltage is set to, $V_{REF} = 7.75 \text{ V}$. Determine what will be the number of clock pulses counted by the counter and the binary form of the digital output, when the analog input voltage is $V_{IN} = -3.15 \text{ V}$, and when it is -1.25 V ?

15
(CO2,
PO2)

[solution
is after
the below
concept
stuff]

First, for single slope ADC →



Ramp generator — gives V_r as in the graph.

$$\frac{V_{ref} - 0}{R} = C \cdot \frac{d}{dt}(0 - V_r), i_c = C \cdot \frac{dv}{dt}$$

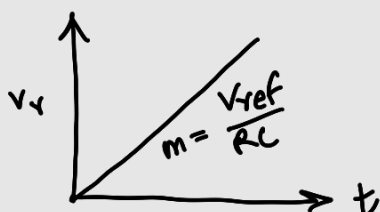
i_R i_C

$$-\frac{V_{ref}}{RC} = \frac{d}{dt} V_r$$

$$V_r = -\frac{1}{RC} \int V_{ref} dt$$

$$\therefore V_r = -\frac{V_{ref} \times t}{RC} \rightarrow x$$

y_2

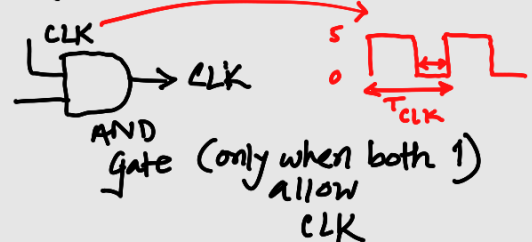


omit $\ominus$ sign by $-V_{ref}$ and $+slope$

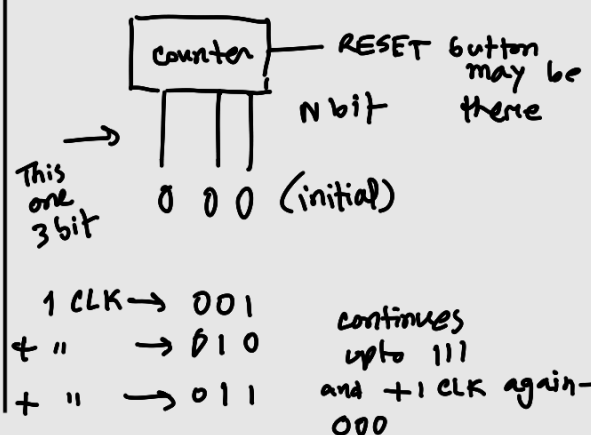
comparator →

high → 5 V

low



counter →



Latches $\rightarrow$

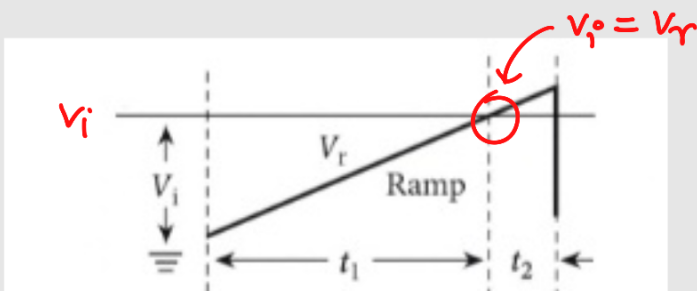
Enable pin $\rightarrow$ high,
values will be stored

Control logic $\rightarrow$

doing reset, enable, clear
based on comparator signal

Working mechanism

- Initially all values are cleared
- V_i given to analog input
- Ramp generator starts increasing V_r
- As long as $V_i > V_r$, CLK sent to counter
- When



counter stops because CLK stops, upto then value stored

Control logic $\rightarrow$ gives Enable and Counter value stored, reset counter, reset Ramp generator

Process complete.

Latch value is the correct representation.

How do we know if correct or not?

$$\begin{aligned} V_r(t_1) &= -\frac{1}{RC} \int_0^{t_1} (-V_{ref}) dt \\ &= \frac{V_{ref} \times t_1}{RC} \\ &= V_i \end{aligned}$$

$$t_1 = n \times T_{CLK}$$

(counted
upto n)

Let, $n=14$

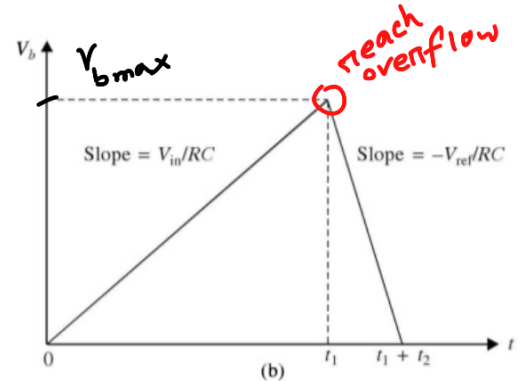
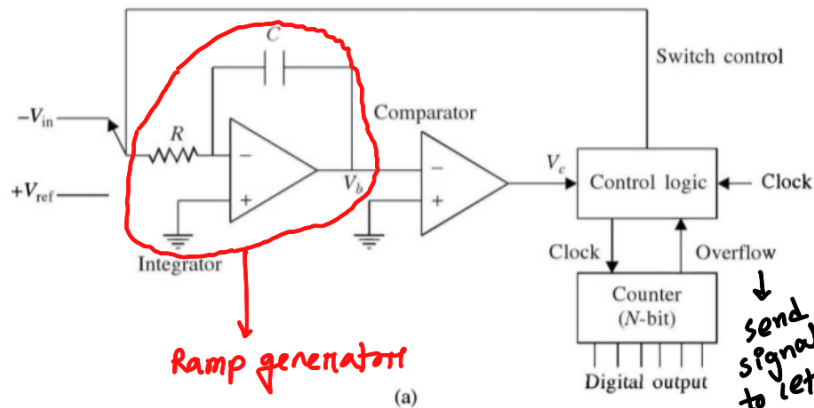
$$V_i = \frac{V_{ref} \times n \times T_{CLK}}{RC}, \quad V_i \propto n$$

↓ so obviously this
will be equivalent.

Limitations →

$$n = \frac{V_i \times \cancel{RC}}{V_{ref} \times T_{CLK}} \rightarrow \text{dependent on this, Temp and environment can affect, so noise can be there.}$$

Dual slope ADC



↓ send signal to let know max value reached

But sometimes controlled by V_{in} and sometimes by V_{ref}

$$V_b = -\frac{1}{RC} \int_0^t (-V_{in}) dt \quad [\text{when } V_{in} \text{ given}]$$

Initially $V_b = 0$, counter $\rightarrow$ reset

$$V_b = \frac{V_{in} \times t}{RC}$$

Comparator $\rightarrow \ominus$ higher than $\oplus$, so logic low,

Control logic $\rightarrow$ CLK should pass and go to counter.

Counter continues upto max, reach overflow $\rightarrow$ switch control changes to V_{ref} , counter reset

$$V_{bmax} = \frac{V_{in} \times t_1}{RC}$$

$$\text{after} \rightarrow V_b = -\frac{V_{ref} \times t}{RC}$$

slope negative

Process complete $\leftarrow$ high signal sent to control logic
CLK stop

V_b decrease up until 0, may cross, $\oplus$ higher than $\ominus$ in comparator

$$V_{b2} = -\frac{1}{RC} \int_{t_1}^{t_1+t_2} V_{ref} dt = -\frac{V_{ref} \times t_2}{RC}$$

$$0 \rightarrow V_{bmax} \rightarrow 0$$

$$\frac{V_{in} \times t_1}{RC} = \frac{V_{ref} \times t_2}{RC}$$

$$V_{in} \times t_1 = V_{ref} \times t_2$$

$$V_{in} \times 2^N = V_{ref} \times n$$

$$n = \frac{V_{in} \times 2^N}{V_{ref}}$$

$$n \propto V_{in} \quad (\text{equivalent}) \quad \checkmark$$

$$t_1 = 2^N \times T_{CLK}$$

Counter N bit
can count upto

$$t_2 = n \times T_{CLK}$$

num
of count

number of clock pulses counted = 2^N (When $2^N - 1$ counted, after, 2^N , overflow, then the counter stops, by the 2^N counts happened)

Solution

[circuit diagram]

[Graph] $\rightarrow t_1, t_2$

Formulae $\frac{V_{in} \times t_1}{RC} = \frac{V_{ref} \times t_2}{RC}$ (From graph we can say that)

$$t_1 = 2^N \times T_{CLK}$$

$$t_2 = n \times T_{CLK}$$

(N bits)

$$\text{total conversion time} = t_1 + t_2$$

dual slope ADC

$$T_{CLK} = \frac{1}{10^6} \text{ s}$$

$$V_{ref} = 7.75 \text{ V}$$

$$N = 5 \text{ bit}$$

$$V_{in} = -3.15 \text{ V}$$

$$t_1 = \frac{3.15}{7.75} \times 2^5$$
$$= 13.00 \text{ s}$$

$$t_2 = 5.2839 \text{ s}$$

$$\left. \begin{array}{l} t_1 + t_2 \\ T_{CLK} \end{array} \right\} = 2^n$$
$$n = 24$$

binary $\rightarrow$ 11000

- b) If a reference voltage of 10 V is applied to a 3-bit Flash ADC, draw the complete circuit diagram and truth table showing the voltage ranges, output from the comparators and output from the priority encoder.

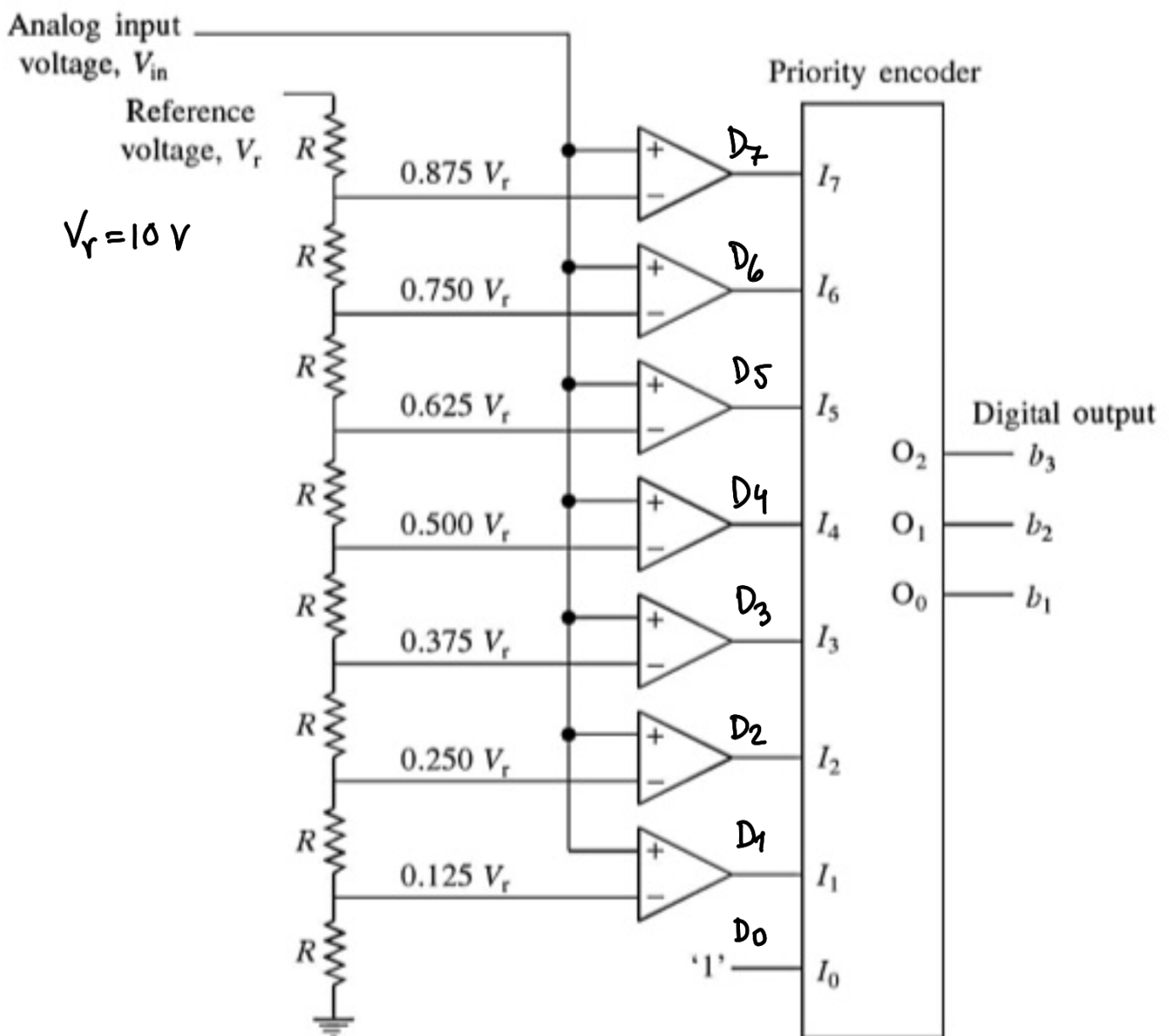
8

$$V_{\text{ref}} = 10 \text{ V}$$

$$N = 3$$

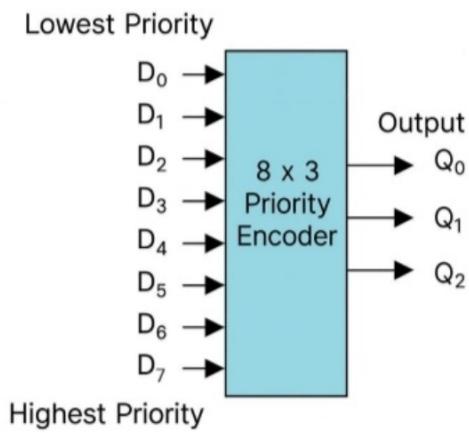
$$L = 2^3 = 8$$

$$q = \frac{10 - 0}{8} = 1.25 \text{ V}$$



When $V_{in} >$ (voltage given to $\leftarrow$) of comparator)
comparator gives logic high
and when $V_{in} <$ (voltage given to $\rightarrow$)
comparator gives logic low

Output States	Discrete Voltage Ranges (V)	Binary Code
0	0.00 - 1.25	000
1	1.25 - 2.50	001
2	2.50 - 3.75	010
3	3.75 - 5.00	011
4	5.00 - 6.25	100
5	6.25 - 7.50	101
6	7.50 - 8.75	110
7	8.75 - 10.0	111



Inputs								Outputs		
D7	D6	D5	D4	D3	D2	D1	D0	Q2	Q1	Q0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	x	0	0	1
0	0	0	0	0	1	x	x	0	1	0
0	0	0	0	1	x	x	x	0	1	1
0	0	0	1	x	x	x	x	1	0	0
0	0	1	x	x	x	x	x	1	0	1
0	1	x	x	x	x	x	x	1	1	0
1	x	x	x	x	x	x	x	1	1	1

Fig: 8 x 3 Priority Encoder

2. a) The output signal of the sample and hold circuit of Question 1(b) is provided as one of the input to the comparators of a 4-bit Flash ADC with a reference voltage of 20V. Sketch the circuit diagram of this 4-bit Flash ADC. Describe the operating principle of this circuit. Find the equivalent digital form of the signal for the entire duration of 4ms.

14
(CO2,
PO2)

4x16 priority encoder needed

- d) If the analog signal to be quantized (unipolar quantization with 4 bits) has a range from 0 V to 10 V. Determine: (i) number of quantization levels, (ii) quantization step (resolution), (iii) quantization level when the analogue voltage is 7.4 V, and its binary code. (iv) quantization error when the analog input is 7.4 V. (v) quantization level, binary code, and quantization error when the analog voltage is the third sample of the signal $x(t) = |3 \sin(2\pi t)|$ sampled at 16 sample/sec. (note that 1st sample is at $n = 0$)

[6]
CO6,
PO1

$$i) 2^4 = 16$$

$$ii) \frac{10-0}{16} = 0.625$$

$$iii) V_m = 7.4$$

$$n \times 0.625 = 7.4$$

$$n = 11.84 \approx 12$$

$$12 \times 0.625 = 7.5$$

6.875 - 7.5 range for 12

$$11 \times 0.625 = 6.875$$

6.25 - 6.875 " " 11

So, binary code for 12 $\rightarrow$ 1100 is the code for 7.4

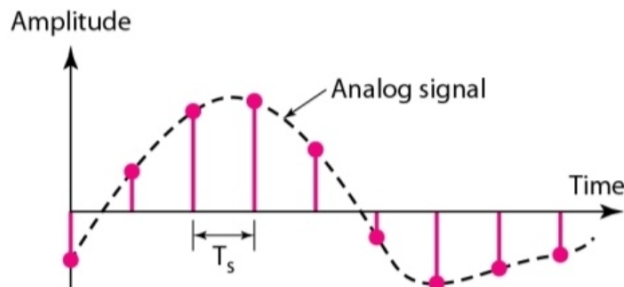
iv) quantization error for 7.4 v

$$q = 7.4 - 7.5$$

$$= -0.1 \text{ V}$$

c) Explain the operation of a sample and hold (S/H) circuit. Draw the generic block diagram of S/H circuit. Assuming sinusoidal signal as input, draw the output from the switching gate, sampled output and final version of the signal from the S/H circuit.

7

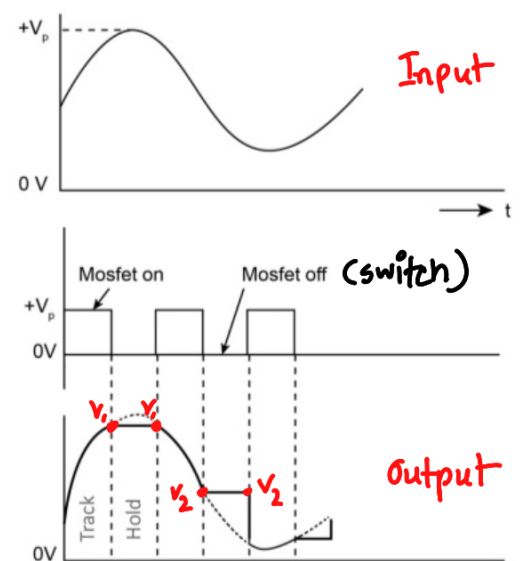
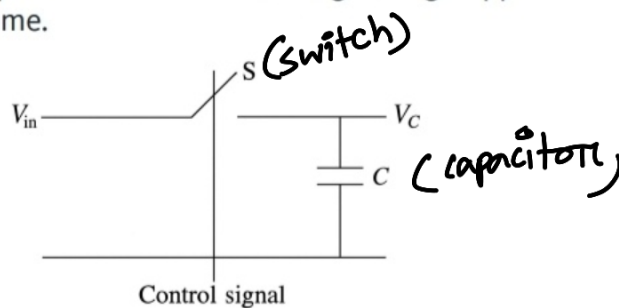


Sampling is the process of converting an analog signal, which is **continuous in both time and amplitude**, into a discrete-time signal, which is **discrete in time** but retains a continuous range of amplitude values.

> The sampling interval T_s must be selected carefully to ensure that the discrete-time representation accurately preserves the information content of the original analog signal

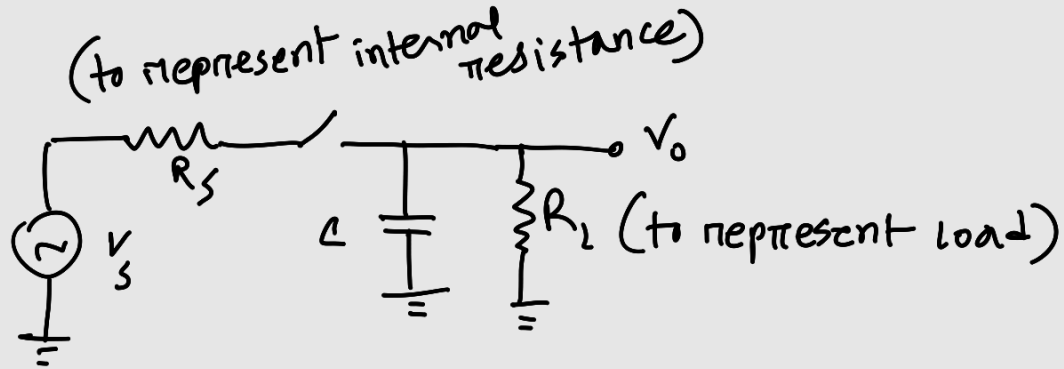
In most ADCs, the input signal must remain constant for a short period before conversion is complete. A sample and hold circuit maintains the input at a **fixed level** during this time, ensuring that conversion accuracy is not affected by signal changes.

When **S** is closed, V_c tracks the applied voltage V_{in} . When **S** is opened, capacitor retains the analog voltage applied at that instant of time.



Value of sample is held for a time $\rightarrow$ why? different steps involved, sampling, quantization, encoding. if momentarily sample and don't hold it, process not completed

When switch closed, output tracks input
 when opened, capacitor which was storing the voltage will give that value (v_1, v_2) to V_c

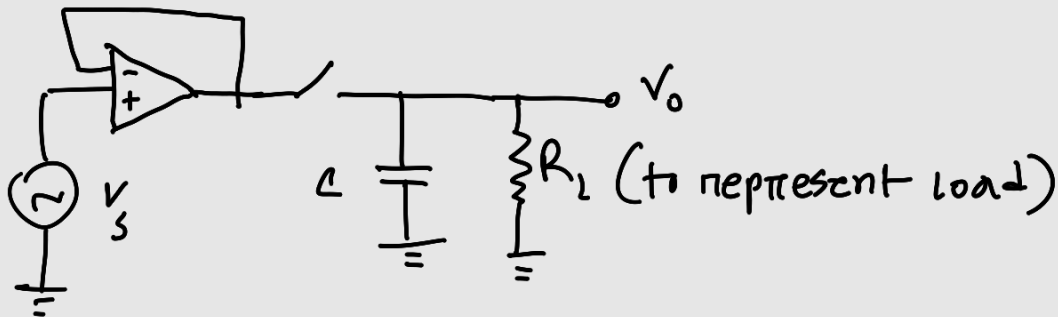


C takes time to charge $T_1 = R_s \times C$

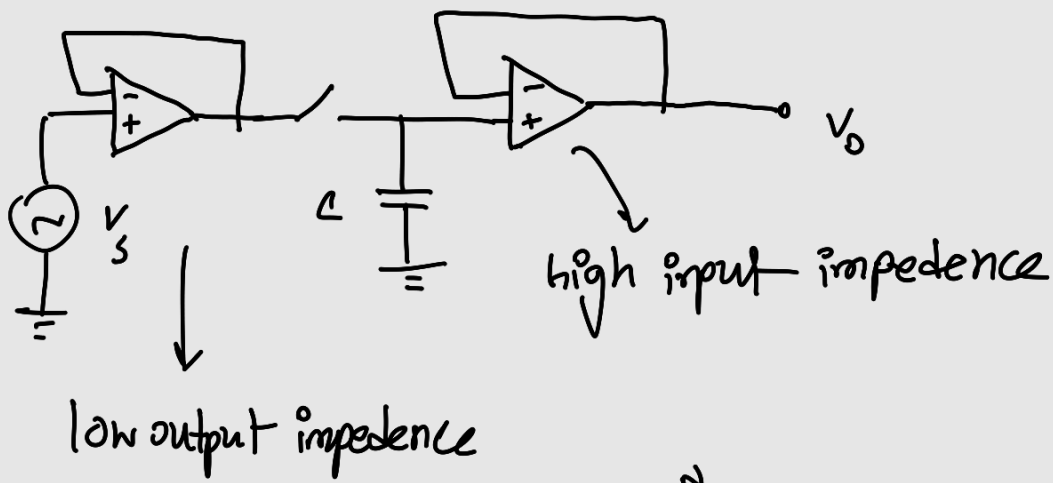
We want no lagging ideally, $5T_1$ (fully charged) instant charge
 So, R_s to be as low as possible

To discharge $T_2 = R_L \times C$ (as high as possible) $R_L \uparrow$

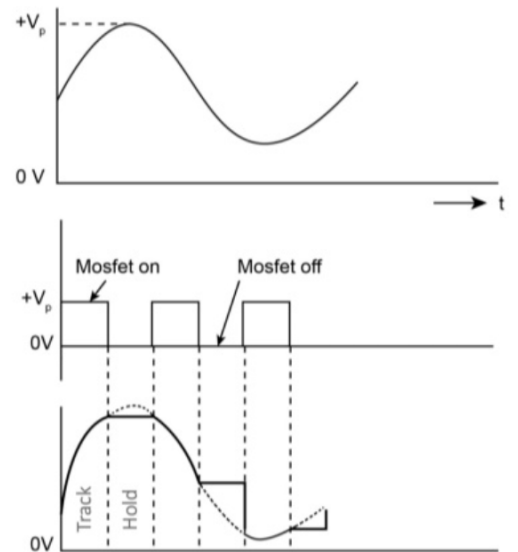
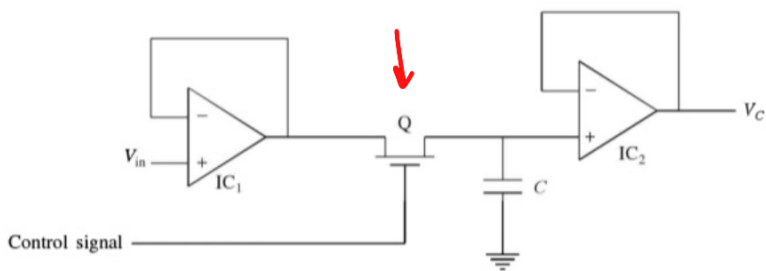
Modification -

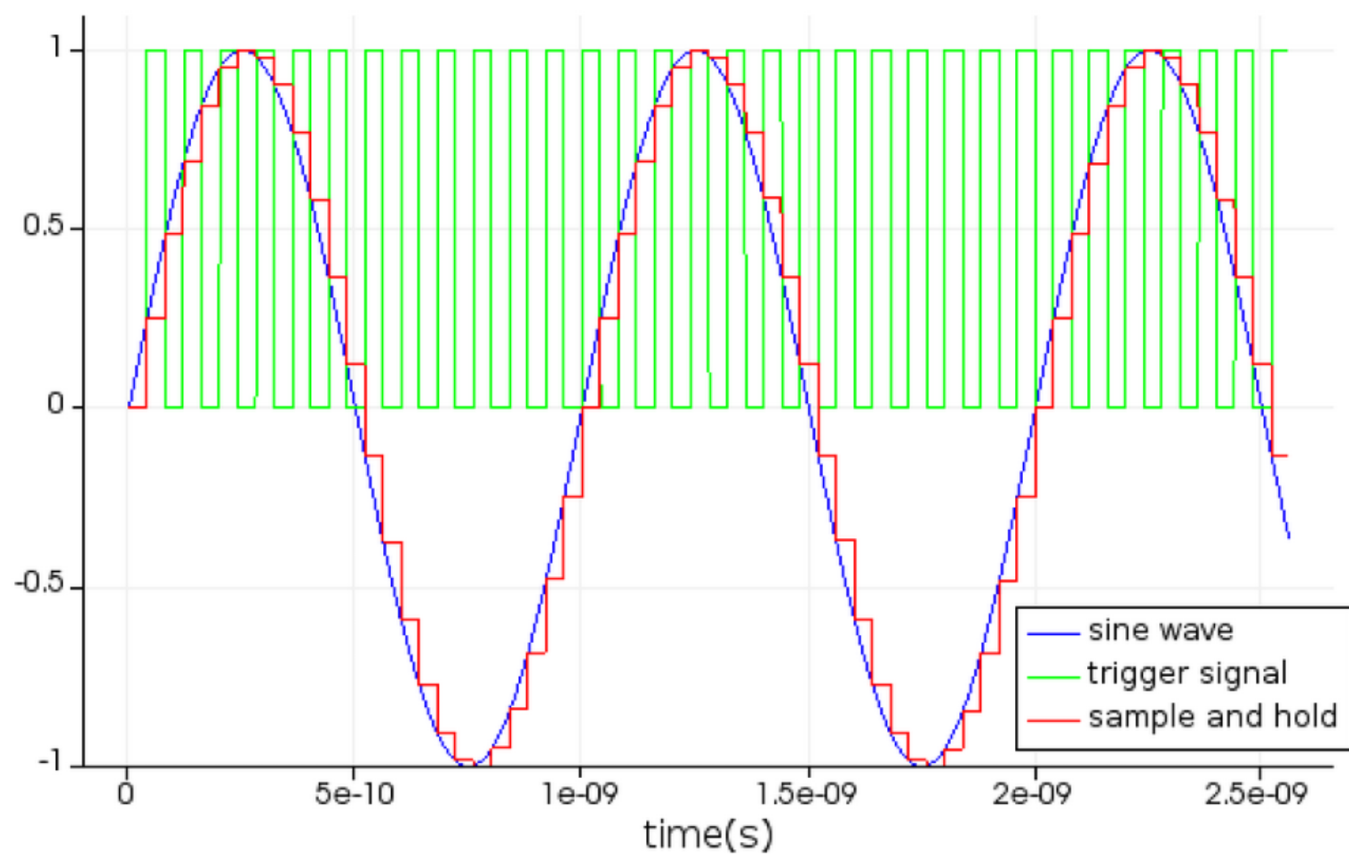


Op-amp buffer replaces R_s
 Why? $\rightarrow$ input impedance $\uparrow$ (needed)
 output $\downarrow$
 effective resistance becomes quite low



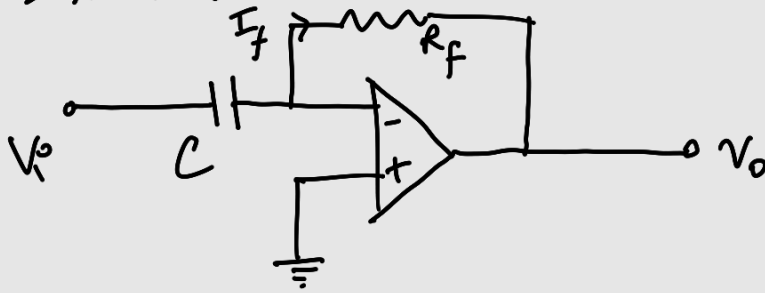
n MOSFET used
control signal high $\rightarrow$ ON
low $\rightarrow$ OFF





d) With relevant equations and diagrams explain the operation of OpAmp as Differentiator and Integrator. 2+1

Differentiator -



0V given here

output voltage proportional to input's rate of change w.r.t time, i.e. differentiation

$$I_{in} = I_f = -\frac{V_o}{R_f}$$

(through C)

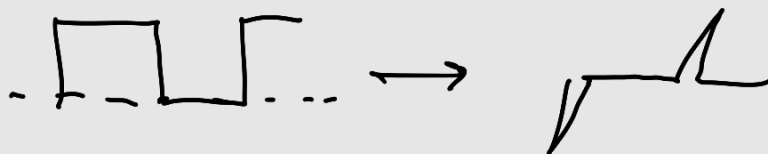
$$\frac{d}{dt} Q = C \cdot \frac{d}{dt} V_{in}$$

$$I_{in} = C \cdot \frac{d}{dt} V_{in} = I_f$$

$$-\frac{V_o}{R_f} = C \cdot \frac{d}{dt} V_{in}$$

$$V_o = -R_f \cdot C \cdot \frac{d}{dt} V_{in}$$

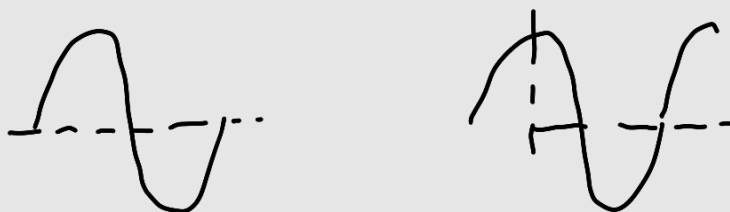
square wave $\rightarrow$ spikes



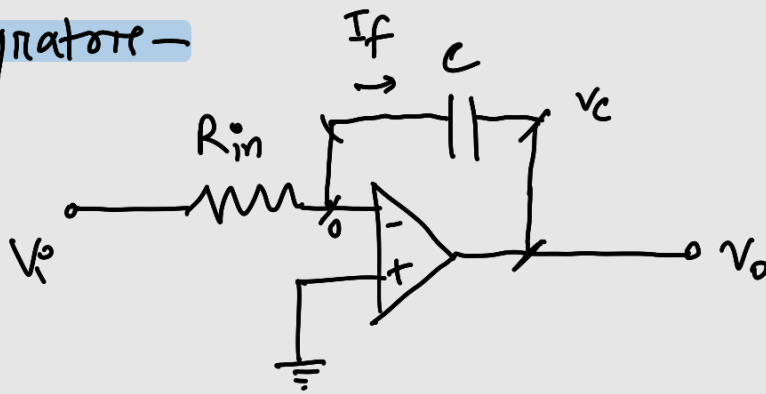
triangular wave $\rightarrow$ rectangular wave



Sine wave $\rightarrow$ cosine



Integrator -



$$V_o = -\frac{1}{R_{in} C} \int_0^t V_{in} dt$$

Inverting amplifier whose output
voltage $\propto$ negative integral of input
voltage simulating math integration

Infinite – Input impedance is the ratio of input voltage to input current and is assumed to be infinite to prevent any current flowing from the source supply into the amplifiers input circuitry ($I_{IN} = 0$). Real op-amps have input leakage currents from a few pico-amps to a few milli-amps.

High Input Impedance of an Op-Amp

Definition:

The input impedance of an op-amp is the resistance “seen” by a signal source connected to its input terminals. A high input impedance means the op-amp draws very little current from the source.

Significance / Why it matters

Minimal Loading of the Source

High input impedance prevents the op-amp from loading the preceding circuit, i.e., it does not significantly draw current.

Example: If a sensor outputs a small current, a low-impedance load could alter the sensor's voltage, causing measurement errors. A high-impedance op-amp avoids this.

Accurate Voltage Measurement

Since the op-amp draws negligible current, the voltage at its input is nearly identical to the source voltage.

Important in voltage follower / buffer configurations, where the op-amp copies the voltage without affecting the circuit.

significance of high input impedance of op-amps

b)	Sketch the circuit diagram of 3 input OR gate using Diode Logic and describe it's working principle.	7 (CO1, PO1)
6. a)	Sketch the circuit diagram of 3-input NAND gate using Diode Transistor Logic (DTL) and describe the operation of the circuit for two possible cases: (i) when all the inputs are at logical '1' state (ii) when any one of the inputs is at logical '0' state (Assume reasonable values of different components for the analysis)	12 (CO1, PO1)
b)	With proper reasoning, describe how TTL gates overcome the speed limitation of DTL gates.	10 (CO1, PO1)
c)	Describe the difference between depletion type MOSFET and enhancement type MOSFET.	3 (CO1, PO1)
c)	Sketch the circuit diagram of a 3-input TTL NAND gate with active pull-up. Specify the role of each transistor in the circuit and describe the overall operation of the circuit.	[10] (CO1, PO1)
5. a)	Define enhancement type MOSFET. Describe the different regions of operation of an enhancement type nMOSFET and draw it's typical IV characteristics graph.	12 (CO1, PO1)
b)	Define rise time, fall time and propagation delay related to the operation of logic gates.	6 (CO1, PO1)
c)	Sketch the circuit diagram of 3 input AND gate using Diode Logic and describe it's working principle.	7 (CO1, PO1)

3 input DL OR

3 input DTL
NAND

3 input TTL

3 input DL AND

- b) In Fig. 1, gate G1 is a 3-input DTL NAND gate that has a fan-out equal to N (output of G1 is driving one of the inputs of N other similar 3-input NAND gates). If all the inputs of gate G1 is at logical HIGH state, determine the approximate value of N. [Here, $\beta = 50$ and $\sigma = 0.85$] [10] (CO2, PO2)

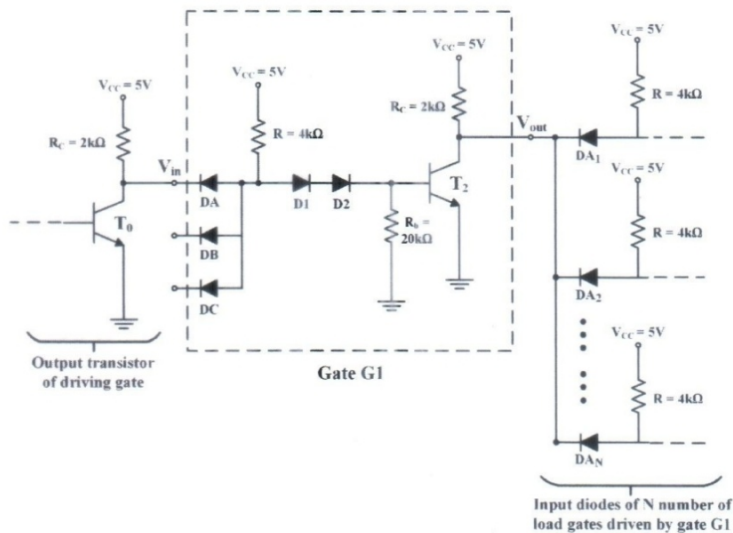


Fig. 1

- b) In Fig. 2, an RTL-based NOT gate is working as the driver gate that has a Fan-out equal to N. Calculate the minimum voltage at the input side of each NOT gate guaranteed to be recognized as logic HIGH. Determine the value of N. What will be the effect on Fan-out if a noise margin of 0.5V is considered. [Here, $V_{BE}(\text{Sat}) = 0.75\text{V}$, $V_{CE}(\text{Sat}) = 0.2\text{V}$, $\beta = 50$, $\sigma = 0.85$] 12 (CO2, PO2)

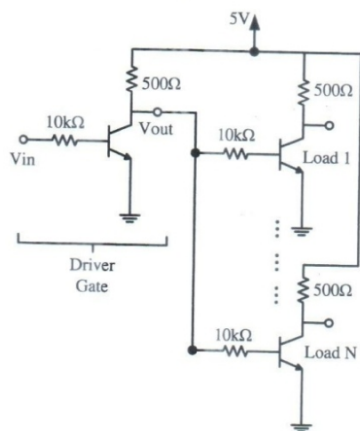


Fig. 2

- b) Describe which type of logic circuit is presented in Fig. 2. For this circuit, sketch the input-output characteristics curve, determine the different input & output voltage levels and noise margins. [Here, $\beta = 50$, $\sigma = 0.85$, $V_{BE}(\text{Cut-in}) = 0.65\text{V}$, $V_{BE}(\text{Sat}) = 0.7\text{V}$, $V_{CE}(\text{Sat}) = 0.2\text{V}$] [13] (CO2, PO2)

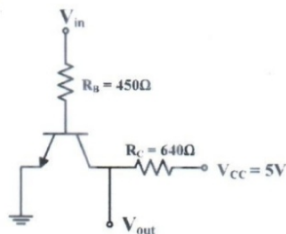


Fig. 2